

Stochastic interacting particles

• Examples

- Ising model: $E = \{\pm 1\}$, discrete time

- Interacting stochastic processes? $dX_t = b(X_t)dt + \sigma(X_t)dW_t$

McKean-Vlasov models

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A McKean-Vlasov model

- $dX_t^i = \sum_{j=1}^N a(X_t^i - X_t^j)dt + \sigma dW_t^i$
- What is the behaviour depending on a, σ, N ?